

SUSTAINABLE TEACHING LABS

PROBLEM DEFINITION

Since the current capacity of first and second year labs and workspaces in J07 and S173 Halliday Lab / Makerspace are at its limit, therefore, new reforms are needed to increase the functional capacity of the lab and workspace while confirming that these areas follow sustainable practices when in use. At the same time, ensuring that the policies laid by the stakeholders is followed at all times while tackling these challenges.

STAKEHOLDERS

The main stakeholders of this project are as follows:

- The Faculty of Aeronautical, Mechanical and Mechatronics Engineering (AMME)
- Prof. Matthew Cleary
- Dr Shuying Wu
- The University of Sydney
- Government
- Students

ENGINEERING REQUIREMENTS

- All lab equipment must adhere to current 2024 safety protocols as outlined by the University of Sydney's policies.
- Reduce Scope 1 and 2 emissions and material wastage from the labs by 5 to 10% in the first year (i.e. 2024) of implementing the new spaces/ideas.
- Budget allocated by the University is to be followed at all times as per the *Building Projects Approval and Management Policy 2014*.
- New spaces must be equipped with appropriate safety measures as per safety standards laid by the University of Sydney.
- Special waste disposal unit must be installed to improve proper disposal of organic, hazardous and plastic waste in each teaching lab.
- Workplace health and safety (WHS) policies and procedures must be implemented throughout the construction and operation of labs.
- Any renovations made must not compromise the heritage of the J07 building.
- All equipment and furniture must be accessible to students with diverse physical abilities and neurodiversity.
- High-quality demonstrators and tutors must be selected to teach in the session as per the standards set by University of Sydney.
- Increasing Functional Capacity of the current teaching labs and workspaces.

OUR PROPOSAL

A multi-disciplinary approach is needed to ensure that we optimise the maximum out of the current capacity and space constraints. We plan to do this by remodeling the labs and workspaces by introducing foldable whiteboards & furniture, flexible and movable furniture, maintaining the lab equipments as per the current safety standards,tactile improvements in the labs and workspaces for increased accessibility and ADHD-friendly resources - without damaging the original structure or heritage of the building.

We would also like to implement improved Heating Ventilation and Air Conditioning (HVAC) Systems along with motion sensed lighting, special waste disposal units, automatic alarm system and thermal fire detection in the labs and workspaces for increased safety of the staff and students and sustainability. The Faculty of AMME is also trying their utmost to reduce the emissions to net-zero by 2030 under the University's Net-Zero Initiative. This is aligned with the legal, policy and environmental contexts of the stakeholders. Furthermore, addition of competitive staff who know how to use the equipment and collaborative workstations will make the sessions more interactive, which is aligned with the social context.

STRENGTHS

- All the steps in our plan is made in accordance with the University's policies.
- Increased safety for the staff and students.
- Increased accessibility to various types of students, hence promoting inclusivity and diversity.
- Reduced emissions and energy consumption with installation of new infrastructure.

LIMITATIONS

- The budget for remodeling the labs and workspaces may or may not get approved by the University.
- Implementing the new system and remodeling all the labs and workspaces will require considerable logistics and time.
- There still may be limitations on the extent of optimisation of the current labs/workspaces.
- Regular maintainence of motion sensed lighting, HVAC System, etc. is required, adding to the operational costs.

COST ESTIMATIONS

Item Name	Quantity	Price per Unit (\$)	Total Price (\$)
HVAC System	2	10,000	20,000
Motion Sensor Lighting	7	56	392
Foldable Whiteboard Desk	2	800	1,600
Custom Tactile Signage	2	51	102
Bone Conduction Headphones	10	50	500
Hired Demonstrators	4	85,000	340,000
TOTAL COST			362,594

REFERENCES



SCAN ME

QUANTITATIVE FEASIBILITY

FINANCIAL ANALYSIS: The plan is financially feasible for the remodeling and installation of new infrastructure, considering the spaces can be remodeled in stages as the faculty gets funding from the University. After the remodeling and installations are done, regular maintenance of systems like HVAC, and automatic alarm systems will be required. These will add to the existing operational costs, though, these systems will help in reducing emissions and energy consumption, hence, improving sustainability and safety. Figure 1 showcases the current emission sources of university labs and these will be greatly decreased through our improvements. Moreover, the cost table shown reflects the research conducted to identify the feasibility of this project.

ORGANIZATIONAL ANALYSIS: All the steps of the proposal are aligned with stakeholders' interests which include making the best out of the current space and capacity constraints, ensuring that the labs and workspaces are sustainable and increasing the accessibility for neuro-diverse and physical-diverse students.

RISK ASSESSMENT: There is a risk in the failure of HVAC and other advanced/automatic systems, and health risks during the remodeling of the labs and workspaces. Additionally, alternative spaces might be required for the students to attend their classes in-semester until all the work is done.

Overall, this proposal for the remodelling of the spaces seems viable after going through all the points discussed above.

LABORATORY CARBON EMISSIONS

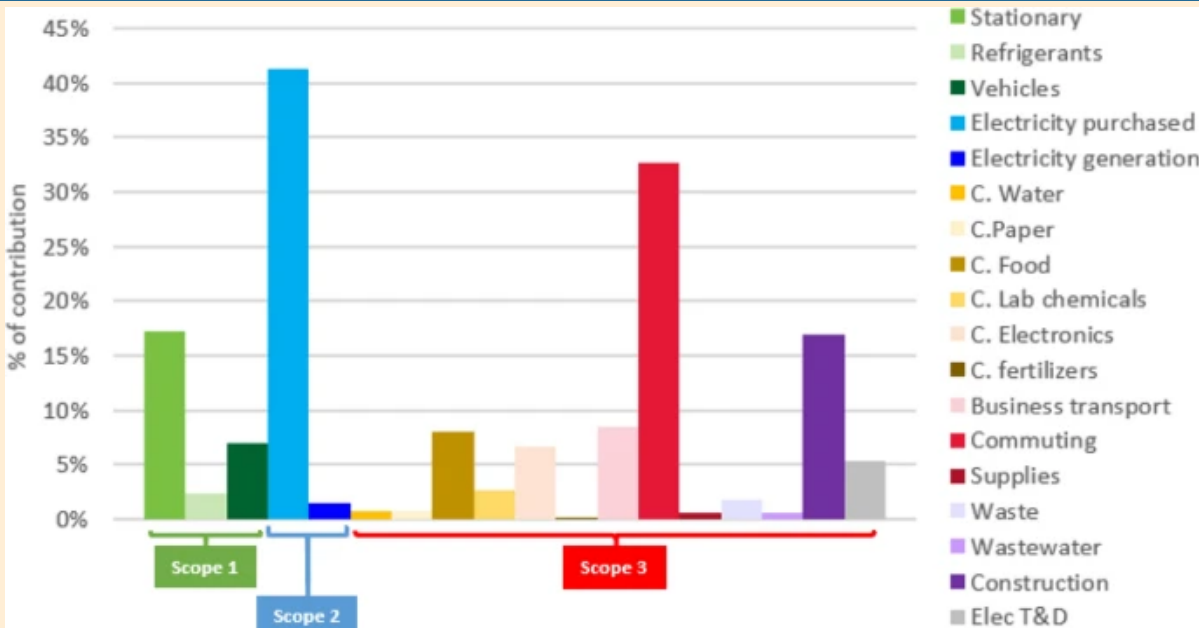


Fig. 1 - Contribution of emission sources from University Labs and classrooms. (Valls-Val & Bovea, 2021)